

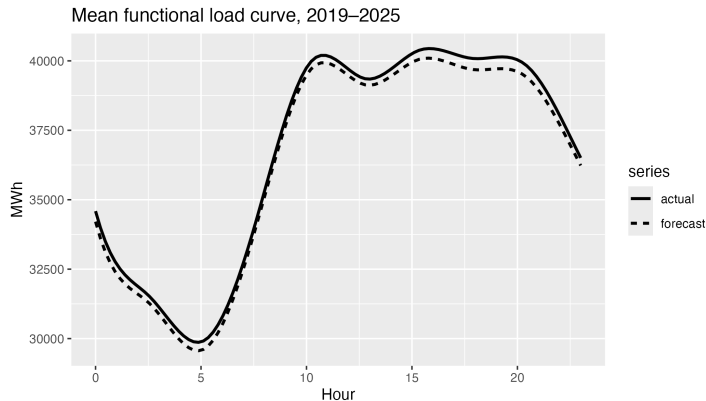
Functional Forecasting of Turkish Electricity Load Curves

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Thesis Proposal

June 5, 2026

The Forecasting Object: A Daily Load Curve



$$Y_d(h), \quad h = 0, \dots, 23$$

- A single **curve**, not 24 scalars
- Dual-peak shape is structurally stable across all days
- Solid = actual; dashed = official forecast
- **Goal:** forecast tomorrow's full 24-hour profile

The official forecast (dashed) lies *systematically below* the actual — persistent underprediction at every hour.

The Load Curve Is an Economic Story

Segment	Hours	Economic driver	vs. official	Why functional analysis fits:
Overnight trough	0–5	Factories + infra; stable, weekly-seasonal	Wins 40–60%	These 24 observations are one coherent trajectory shaped by overlapping behavioral rhythms.
Morning ramp	6–8	Demand doubles; ramp <i>timing</i> is temperature-driven	Loses (up to 24%)	
Midday plateau	9–16	Industrial & commercial peak; weekday-driven	Wins 20–35%	The morning ramp, midday plateau, and evening peak are <i>features of the whole curve</i> — not of individual hours in isolation. FDA works with these shape features directly.
Evening surge	17–19	Residential peak; AC/heating at max; largest official bias	Mixed	
Night wind-down	20–23	Smooth decline toward trough	Wins 17–20%	Red rows are transition segments where demand changes fastest and forecasting is hardest. Losses there motivate the residual correction step.

Why Accuracy Matters

Key users (post-2013 liberalisation):

- **TEİAŞ** (transmission system operator)
unit commitment, reserve scheduling
- **Generators**
day-ahead market bids on EPIAS
- **Large consumers**
real-time imbalance penalties

Why does the official underpredict?

Most likely: **model staleness**. Official systems require regulatory approval to update; a model calibrated on older demand patterns underpredicts during structural demand growth. Our rolling-origin re-trains before every forecast step — 2024 bias: **34 MWh** vs. official **585 MWh**.

Systematic underprediction: official TEİAŞ forecast

Year	Official bias (MWh/hour)	Our bias (MWh/hour)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Average	+467	+103

Positive bias = forecast < actual (underprediction).

Research Questions

- ① **Representation.** Can B-spline smoothing and FPCA represent Turkey's daily electricity load curves with negligible information loss? How many functional principal components suffice?
- ② **Forecasting accuracy.** Does a functional VAR model augmented with holiday and derivative features statistically outperform the official TEİAŞ day-ahead forecast?
Evaluated by Diebold-Mariano over 1,096 hold-out days, 2023–2025.
- ③ **Score linearity.** Is the linear OLS score equation adequate, or does the feature–score relationship contain meaningful nonlinear structure?
Tested by replacing OLS with Random Forest in the score equation, holding the FPCA basis, features, and rolling-origin design fixed.

Literature Positioning [to be expanded]

Functional Data Analysis

- Ramsay & Silverman (2005): FDA foundations
- Kokoszka & Reimherr (2017): modern inference
- Hyndman & Ullah (2007): functional forecasting (demographics)

FDA applied to load curves

- *[to be identified in literature review]*

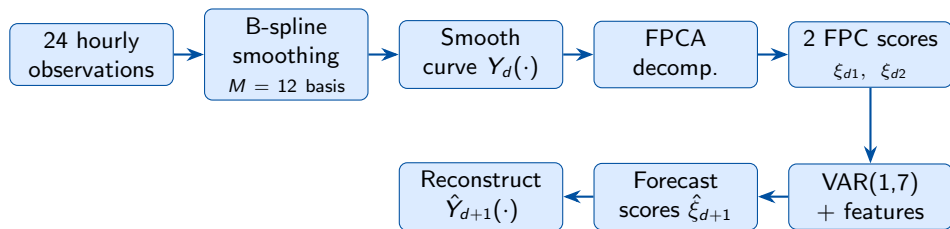
Short-term electricity forecasting

- Classical: ARIMA, exponential smoothing on 24 scalars
- ML: gradient boosting, neural networks
- Xie & Hong (2016): residual simulation and correction

Gap this thesis fills

- No FDA study on Turkey's post-liberalisation market
- Holiday + derivative features for Turkish market structure
- DM testing against the operational benchmark
- Linear score equation validated against nonlinear alternative

From 24 Numbers to a Curve: The FDA Pipeline

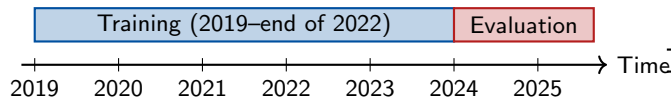


B-splines ($M = 12$)
preserves shape; enables
differentiation of the curve

FPCA ($K = 2$)
2 scores carry 97.3% of all curve
variation

VAR(1,7) by OLS
daily + weekly score dynamics

Data



Rolling-origin: model re-estimated before every forecast day using only past data. Expanding window — appropriate because eigenfunction shapes are stable across sub-periods.

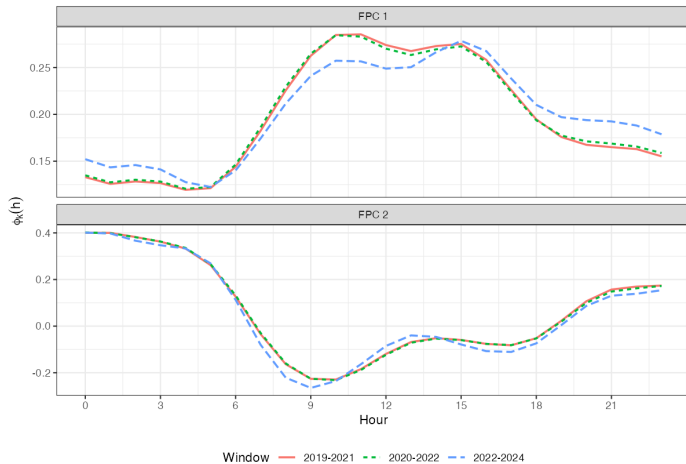
Series	Source
Hourly load (MWh)	EPIAS
Official forecast	TEİAŞ / EPIAS
Holiday calendar	Official gazette

Item	Value
Frequency	Hourly
Coverage	2019–2025
Training curves	≈1,460 days
Evaluation curves	1,096 days
Hourly errors eval	26,304

FPCA: Two Scores Capture the Curve

FPCA eigenfunctions across sub-periods

Overlapping lines = stable eigenfunctions; divergence = structural change



FPC	Variance	Cumulative
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%
4	1.0%	99.4%

FPC 1 — Level

Nearly flat $\Rightarrow$ shifts the whole curve up or down. “How much electricity today.”

FPC 2 — Shape contrast

Positive overnight & evening, negative daytime $\Rightarrow$ peak-vs-plateau contrast.

Stability: sub-period inner products > 0.99

$\Rightarrow$ static FPCA justified empirically

Robustness: $K = 4$ adds $< 1\%$ variance; main results unchanged

Score Forecasting Model

Core dynamics (VAR):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

- A_1 : yesterday's level and shape
- A_7 : same-day-last-week seasonality
- z_d : feature vector, *all predetermined*

Reconstruction:

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\phi}_1(h) \hat{\xi}_{d1} + \hat{\phi}_2(h) \hat{\xi}_{d2}$$

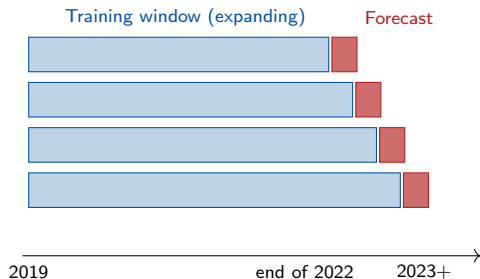
Estimated by OLS, equation-by-equation.

Feature blocks in z_d :

Block	Content
Calendar	Weekend flag, annual sine/cosine
Holiday	Fixed holidays, Ramadan, Bayram, bridge days
Load shape	Yesterday's mean, peak, trough, range
Rolling avg.	7-day, 28-day load averages
Derivative	Morning ramp (hr 6–8), evening ramp, slope volatility

Holiday and **derivative** blocks are critical: without them the model does not beat the official.

Evaluation: Rolling-Origin Design and DM Test



FPCA and VAR re-estimated at every step.
Only data dated $< d$ used to forecast day d .
Mirrors real operational deployment: no information leakage.

Diebold-Mariano test

Loss differential per observation:

$$d_t = |e_t^{\text{ours}}| - |e_t^{\text{official}}|$$

$H_1: \mathbb{E}[d_t] < 0$ (our model has lower loss)

HAC t -test on $\bar{d}$ (Newey-West long-run variance)

+ Harvey–Leybourne–Newbold small-sample correction.

Run **pooled** (26,304 obs.)

+ **hour-by-hour** (24 tests).

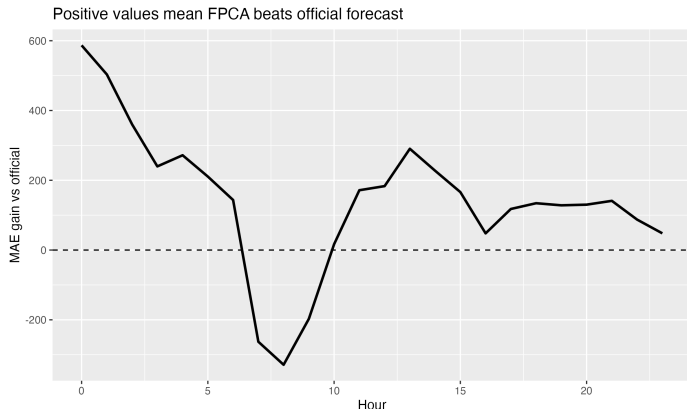
A lower MAE alone is not sufficient evidence.

Main Results: Building the Model Step by Step

Model	MAE	MAPE	vs. official	DM stat	
Seasonal naïve Y_{d-7}	1912	5.13%	-57.6%	+35.2	—
Official (TEİAŞ)	1214	3.15%	ref.	—	
FPCA VAR(1,7) only	1443	3.76%	-18.9%	+22.9	
+ calendar / ramp	1238	3.26%	-2.1%	+2.6	
+ holiday / load-shape	1001	2.63%	+17.5%	-24.9	***
+ derivative features	980	2.56%	+19.2%	-27.4	***
+ residual correction	653	1.70%	+46.2%	-71.2	***

*** $p < 0.001$ vs. official (pooled DM, 26,304 obs.). Positive DM stat = worse than official; negative = better. MAE in MWh/hour. Evaluation: 2023–2025.

Where the Model Wins and Where It Loses



Positive = model beats official (MAE, MWh/hour)

Wins: hours 0–5 and 10–23

Overnight trough and daytime plateau — stable, weekly-seasonal behavior

Losses: hours 6–9

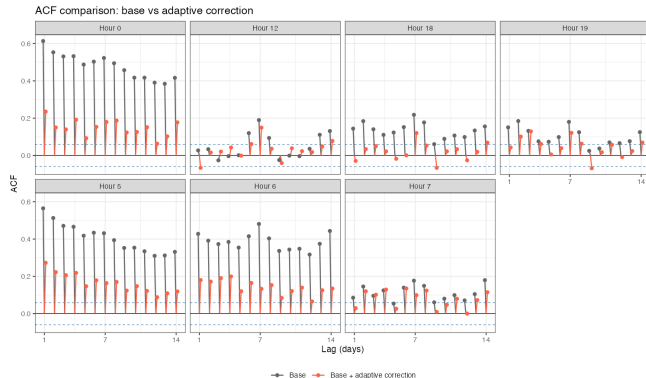
Morning ramp. Demand doubles in 2 hours; ramp timing is temperature-driven. At hour 8, our MAPE loss is **24%**.

DM significant at 18 of 24 hours (5% level).

Persistent, autocorrelated ramp-hour errors motivate the residual correction.

Residual Correction: Exploiting Persistent Errors

Motivation: errors are serially correlated
Ljung-Box rejects $H_0: \rho_1 = \dots = \rho_{14} = 0$ at all 24 hours. LB stat up to 7,227 at hour 6.



$$\hat{Y}_d^{\text{res}}(h) = \hat{Y}_d^{\text{base}}(h) + \lambda_d \hat{r}_d(h)$$

$\hat{r}_d(h)$: correction curve from recent past residuals

λ_d : rolling CV on strictly pre- d data — no leakage

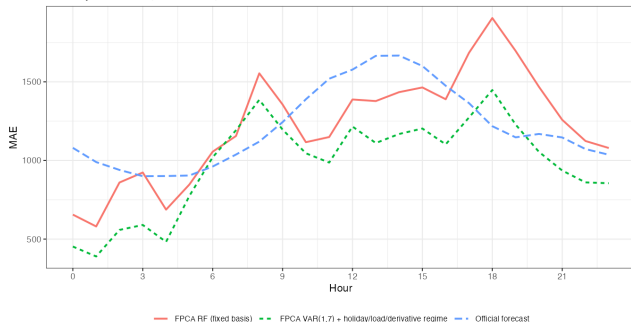
Result:

- +46.2% MAE vs. official (DM stat -71.2)
- Wins at **all 24 hours** including ramp hours

Gray = base model ACF; Orange = after correction

Score Linearity Check: Is OLS Adequate?

MAE by hour: RF (fixed basis) vs OLS vs Official



Design: replace OLS with RF in the score equation.

Same FPCA basis, same features z_d , same rolling-origin design. One RF per FPC score.

Model	MAE	vs. off.	DM
Official	1214	ref.	—
FPCA-RF	1217	−0.3%	n.s.
FPCA-OLS	980	+19.2%	***

Answer to Q3: RF finds no nonlinearity to exploit. FPCA-RF merely ties the official; FPCA-OLS beats both.

OLS's linear constraint is a beneficial regulariser.

The linear score equation is validated.

Direct RF (2025 subsample only, bypasses FPCA):
MAE 718 vs. official 1052 (+31.8%). Different model class; single-year window; not directly comparable to the 2023–2025 main evaluation.

Summary of Contributions

Methodological

- Functional representation of Turkey's daily load curves
B-splines + FPCA; $K = 2$ retains 97.3% of variation; eigenfunction stability confirmed
- Holiday-aware and derivative-augmented feature design for functional VAR
- Residual correction via rolling CV with zero lookahead
- Linear score equation validated against RF:
FPCA-RF ties the official; FPCA-OLS beats both

Empirical

- Base model: **+19.2%** MAE vs. official TEİAŞ
DM $p \approx 10^{-163}$; 1,096 days, 2023–2025
- Residual correction: **+46.2%**; wins at all 24 hours including ramp
- Average bias cut from 467 to 103 MWh/hour
- 2024 bias (structural shock year): 585 MWh official vs. 34 MWh ours

Core claim: Functional representation combined with domain-aware features yields statistically significant, operationally relevant improvement over the official benchmark.
The linear score equation is adequate — no nonlinear extension is needed.

Thesis Road Map

Completed

- ✓ Functional representation (B-splines, FPCA)
- ✓ Rolling-origin evaluation framework
- ✓ VAR(1,7) + feature blocks (holiday, derivative)
- ✓ Residual correction step
- ✓ Diebold-Mariano testing (pooled + hourly)
- ✓ Score linearity check (FPCA-OLS vs FPCA-RF)
- ✓ Eigenfunction stability; $K = 4$ robustness check

Planned

- ☐ Systematic literature review
- ☐ Ramp-hour error diagnostics (write-up)
- ☐ Probabilistic forecasting (prediction intervals)
- ☐ Thesis write-up

References

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